

Machine Learning and Data Science Final Project

Kaitlin Smith

School of Mathematics, Science & Technology, Department of Information Technology

University of the Commonwealth Caribbean

ITT316: Machine Learning and Data Science

Mr. Stefan Watson

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Project Overview

Objective

This project's main objective was to create a reliable machine learning model that could predict used car values across the US on Craigslist. Based on a variety of car parameters, including make, model, year, mileage, condition, and location, this predictive model is intended to provide users, both buyers and sellers, with insights about fair market prices.

Dataset

A specially designed web scraper that periodically gathers information from used car advertisements on Craigslist provided the dataset for this research. With 426,881 records and 25 characteristics, it is a large dataset. Among the features are detailed details like:

- Price: The selling price of the vehicle (target variable).
- Year: Year of manufacture.
- Manufacturer: Brand (e.g., Ford, Toyota).
- Model: Specific model name.
- Condition: Vehicle condition (e.g., new, good, fair, salvage).
- Odometer: Total mileage.
- Geographic Data: Region, state, latitude, and longitude.
- Other Features: Fuel type, transmission, drive, size, type, paint color, and title status.

Methodology

A typical end-to-end data science pipeline was used for the project:

- Data Preprocessing: Handled missing values, eliminated unnecessary columns, and encoded categorical variables to clean up the data.
- In order to comprehend the data and its subtleties, exploratory data analysis (EDA) uses visualized data distributions, feature correlations, and geographic patterns.
- Feature engineering: New features like car_age and mileage_per_year were developed to possibly enhance model performance.
- Model Selection and Training: Using a pipeline with preprocessing steps, a number of regression models, such as Random Forest and Linear Regression, were assessed.
- Model Evaluation: R^2 Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) metrics were used to evaluate the model's performance.
- Interpretation: The most important predictors of car pricing were determined by analyzing feature importance.

Technical Details

Libraries and Tools

- The main programming language is Python.
- For numerical operations and data manipulation, use Pandas and Numpy.
- For making static visualizations, use Seaborn and matplotlib.
- scikit-learn: For modeling (LinearRegression, RandomForestRegressor), evaluation (train_test_split, r2_score, mean_absolute_error), and preprocessing (StandardScaler, OneHotEncoder, SimpleImputer).
- Google Colab: The development environment that made it possible to easily access files on Google Drive and utilize GPU/TPU if necessary.

Data Processing Steps

- Removing Irrelevant Columns: Since they were thought to be unnecessary for modeling, columns like id, url, region_url, VIN, image_url, county, lat, long, and posting_date were removed.
- Managing Missing Values:
 - To guarantee data quality, rows lacking essential characteristics (year, model, gasoline, transmission, and odometer) were eliminated.
 - The string “unknown” was used to substitute missing values in other categorical columns.
- Outlier Removal: Unrealistic values ($\$500 < \text{price} < \$200,000$) were eliminated by filtering prices.

- Encoding: To make categorical characteristics appropriate for machine learning models, they were one-hot encoded using OneHotEncoder inside a ColumnTransformer.
- Scaling: To normalize the ranges of numerical features (year, odometer), StandardScaler was used.

Model Training and Validation

- Pipeline: To ensure that transformations are implemented appropriately and avoid data leaking, a pipeline was utilized to integrate preprocessing and model training.
- Model: A RandomForestRegressor with 50 estimators and a maximum depth of 10 was the main model employed.
- Train-Test Split: A 20% testing set and an 80% training set were created from the data.
- Performance: The resulting model has an RMSE of \$7,786, an MAE of \$4,719, and a R^2 score of 0.7162 on the test set.

Results and Analysis

Key Findings

The Random Forest model's feature importance indicates that the following characteristics are crucial for predicting car prices:

1. year (manufacturing year)
2. odometer (mileage)
3. condition_excellent (If the state is "excellent")
4. condition_like new (If the state is "like new")
5. manufacturer_unknown (an intriguing substitute for missing manufacturer data)

The vehicle's age and mileage are the best indicators of its cost, followed by its manufacturer and condition, according to the model's feature importance plot. Given that newer cars with better condition and less mileage have higher costs, this is consistent with economic intuition. Manufacturer_unknown indicates that many listings do not have manufacturer information, which may indicate a problem with the quality of the data.

Model Performance Evaluation

- The model explains around 71.6% of the variation in car costs, according to its R^2 value of 0.7162. This fit is decent but not ideal.
- The model's forecasts are often wrong by roughly \$4,719 when the Mean Absolute Error (MAE) is \$4,719. Even if this number is reasonable, it may be better.
- The model serves as a reliable baseline and shows that employing a Random Forest model for this task is feasible.

Future Considerations

Potential Enhancements to the Model

1. Hyperparameter tuning: Only a small number of hyperparameters are used in the current model. By determining the ideal `n_estimators`, `max_depth`, `min_samples_split`, etc., a more thorough tuning strategy utilizing `GridSearchCV` or `RandomizedSearchCV` could greatly enhance performance.
2. Feature engineering: To increase predictive capability, more features could be developed:
 - Market Trends: Use metrics for market demand or local economic statistics.
 - Vehicle Density: To determine competition, count the number of comparable postings in the same area.
 - Text Features: If the vehicle description is provided, use Natural Language Processing (NLP) to extract additional information.
3. Model Experimentation: Investigate more sophisticated algorithms that may be able to identify more intricate patterns in the data, such as `XGBoost`, `LightGBM`, or even neural networks.
4. Data Quality: There are a lot of missing values in the dataset (such as size and cylinders). Model accuracy would probably be increased by using more advanced imputation techniques or by improving data collection to minimize missingness.

Application and Deployment

- A framework such as Flask or Streamlit can be used to implement the model as a straightforward web application. A real-time price estimate could be obtained by entering car attributes.
- This technology might be sold to individual buyers and sellers to assist them in making better judgments, or to automobile dealerships for inventory value.

I CERTIFY THAT I HAVE NOT GIVEN OR RECEIVED ANY UNAUTHORIZED ASSISTANCE ON THIS ASSIGNMENT.

AI was used to assist with content structure and grammar correction.

K. Smith